

Practice Test IV (Version 2)

Name
Date
Course

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) A graphing calculator is required.

① After 63 days, a radioactive material retains 18% of its original mass. What is the half life of this radioactive material? Round to the nearest hundredth.

② The population of a strain of E. coli after t hours can be predicted with the equation below.

$$P(t) = \frac{6,000}{1 + 44e^{-0.14t}}$$

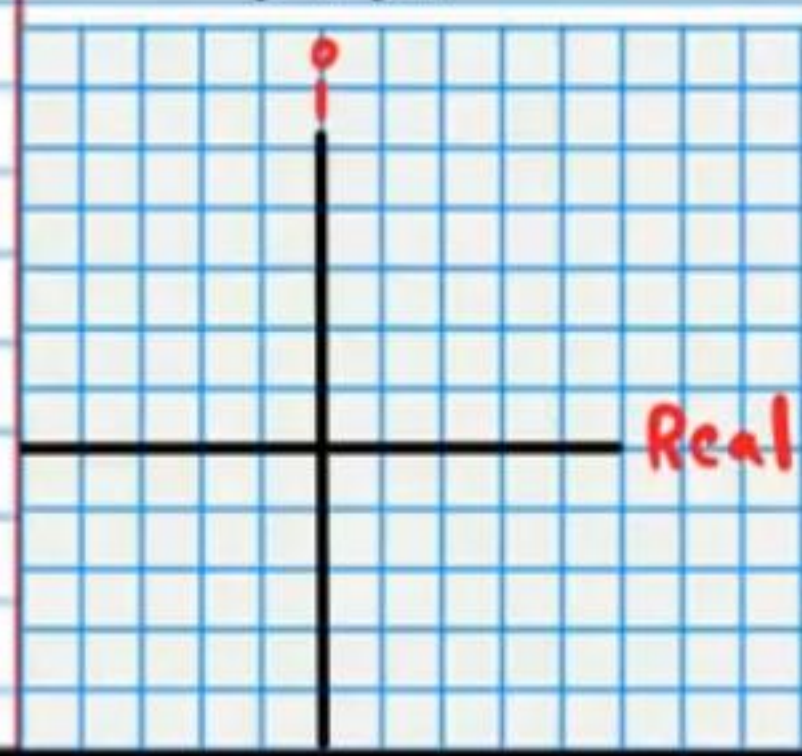
i) What is the carrying capacity of the strain?

ii) What is the initial population? Round to the nearest organism.

iii) What will the population be after 18 hours? Round to the nearest organism.

iv) Graph the following complex number.

$$-3 + 5i$$



③ Solve. Round to the nearest thousandth.

$$17 = 5(e^x) + 2$$

④ Solve. You may leave your answer as an exact expression with an irrational denominator.

$$11 = 7 - 3 \log x$$

⑤ Find the median of the data set below.

17, 15, 10, 12

⑥ i) Darrel went to a business seminar for 4 days. He met 10 people at the seminar. What is the average number of people that he met during his entire stay?

ii) During his time at the seminar, Darrel talked with people for a total of 16 hours. The rest of the time he was sunbathing on the beach. What is the average number of people that he met per day during the time that he talked with people?

⑦ For the function below, find an algebraic expression representing the difference quotient. Use "h" for "change in x". It is not necessary to simplify the expression.

$$f(x) = 1 - 3x$$

⑧ Titanium-45 converts to scandium according to the law of uninhibited decay ($k = -0.00375$). If 32 grams of titanium-45 is present, how many grams would be left after 155 minutes? Assume that k was calculated with units in minutes. Round to the nearest hundredth.

⑨ Find the intensity of sound required to produce the following decibels.

31 dB

⑩ i) If $f(x) = x^2 - 4x - 7$ and $f(b) = -7$, find b .
ii) If $f(x) = -8 + \frac{x}{10}$ and $f(k) = 16$, find k .

⑪ Solve. Round to the nearest thousandth.

$$\frac{\ln x}{2} + 4 = 6$$

⑫ If $g(x) = 3x^2 + 4x$ & $f(x) = 5$, for what value(s) of x does $f(x) = g(x)$?

⑬ Each table below represents a function "f". Write the values of the inverse function "f⁻¹". If "f" is not a one-to-one function, then you must omit values of "f⁻¹" so that it qualifies as a function. To ensure that the functions are perfectly matched inverses, values of "f" may need to be omitted also. You may choose which values to omit. Highlight or circle the omitted values.

i)

X	11	10	18	4	3	X					
f(x)	16	16	16	3	4	f ⁻¹ (x)					

ii)

X	-2	-4	-5	-7	8	X					
f(x)	-3	-1	9	-7	9	f ⁻¹ (x)					

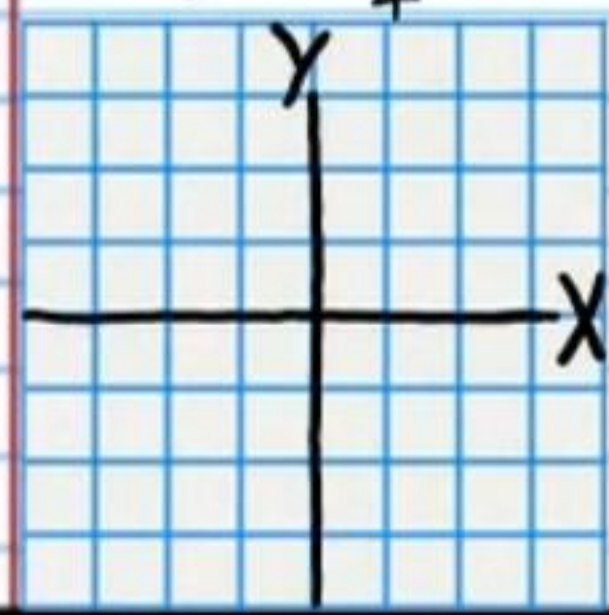
iii) Use your calculator's base "e" logarithm button (LN) and Logarithm Property IX to find the value of the expression below. Round to the nearest ten thousandth.

$$\log_9 12$$

iv) If "f" and "g" are inverse functions, find g(-9) if f(4) = -9.

⑭ i) Graph the following polar equation.

$$\theta = \frac{7\pi}{4}$$



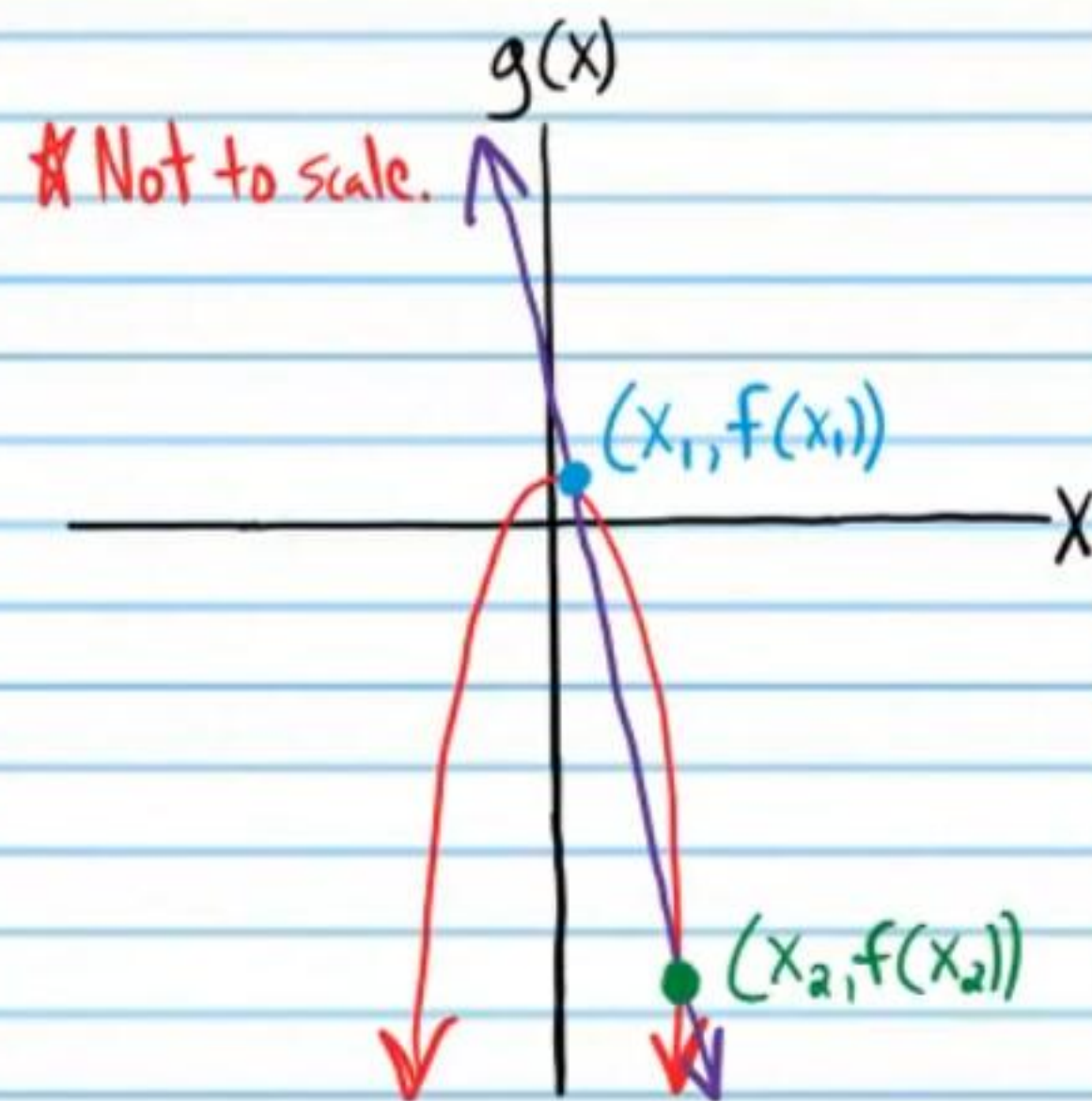
ii) If "f" and "f⁻¹" are inverse functions, find f⁻¹(-100) if f(89) = -100.

⑮ If an investment grows from \$110,000 to \$114,223 at a 3.8% annual interest rate compounded continuously, how much time passed during this growth? Round to the nearest hundredth.

⑯ Are the following functions inverses? Demonstrate algebraically.

$$f(x) = \sqrt[3]{x-2} + 1 \quad \& \quad g(x) = (x-1)^3 + 2$$

⑪ If $g(x) = -2x^2 + 5$, find the average rate of change of the function per unit x from $x=2$ to $x=5$



⑫ If $f(x) = -\left(\frac{x}{4} + 5\right)^2 + 9$ and $f(r) = -10$, find r .

Find an equation of the inverse of each function below.

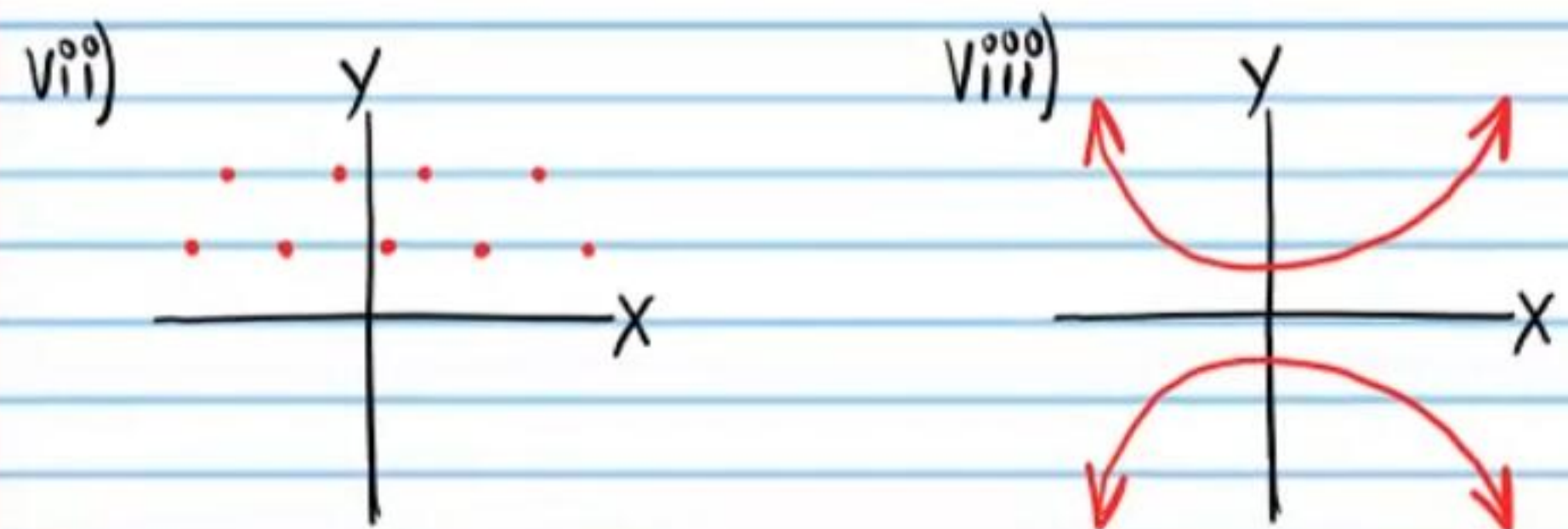
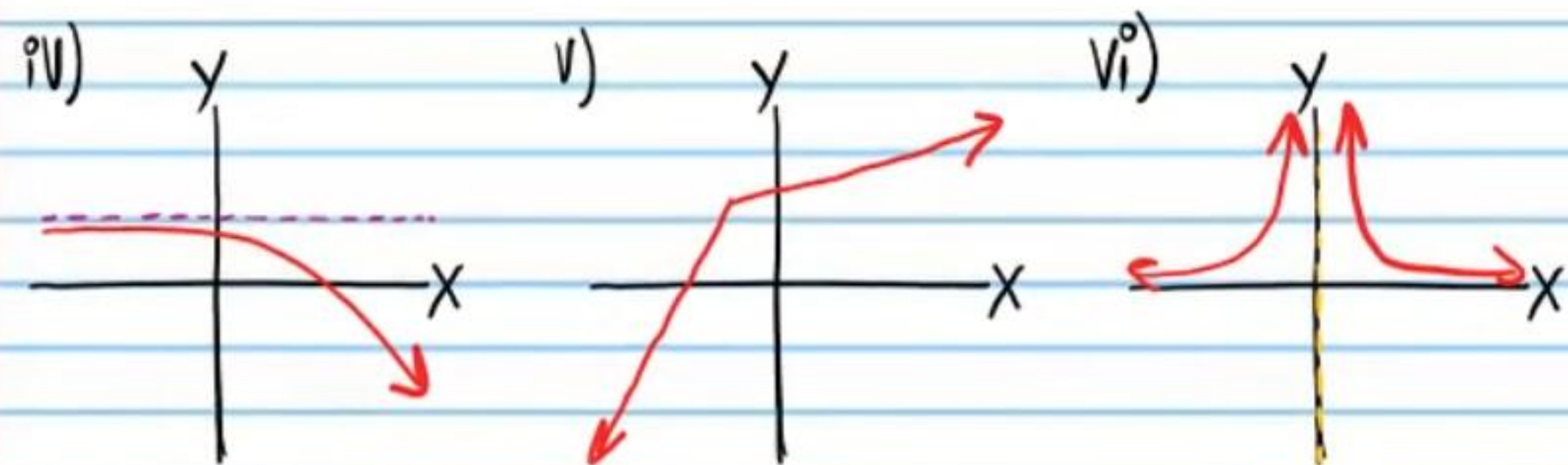
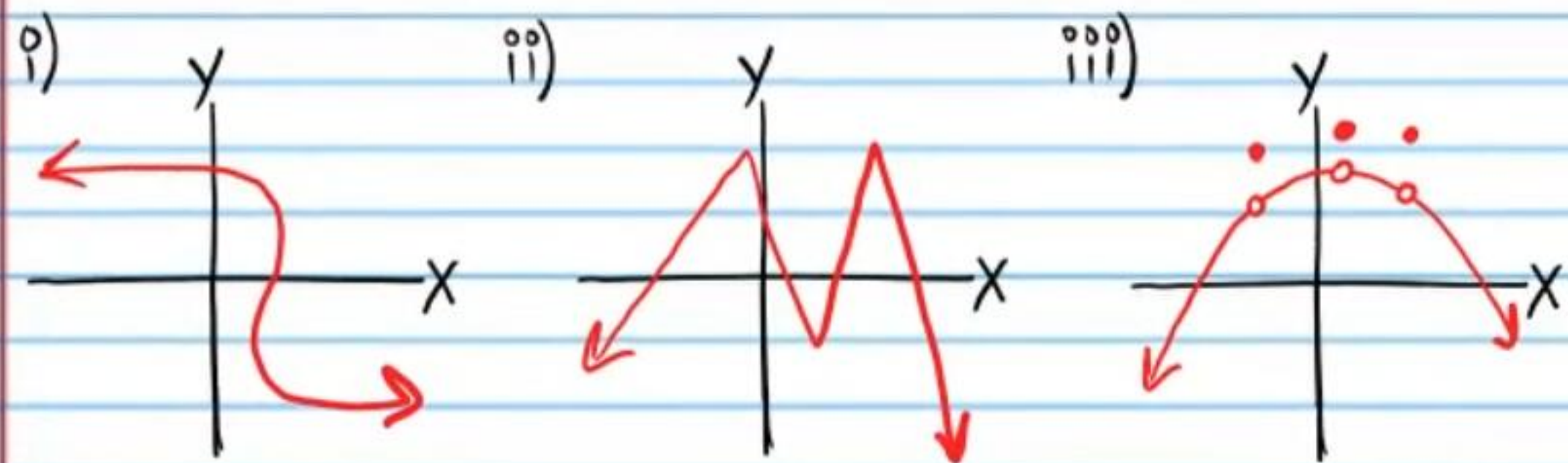
①⑨ $f(x) = \frac{2}{x}$

②⑩ $f(x) = -3(10^x) + 4$

②① Show algebraically that " f " and " f^{-1} " are inverse functions.

$$f(x) = \frac{3}{5x+2} \quad \& \quad f^{-1}(x) = \frac{3}{5x} - \frac{2}{5}$$

22) Relations are written below in the form of graphs. Some of the graphs could represent functions that are one-to-one, some of them could represent functions that are not one-to-one, and some cannot represent functions. Determine the scenario for each graph. Write "1-to-1 function," "function," or "not a function."

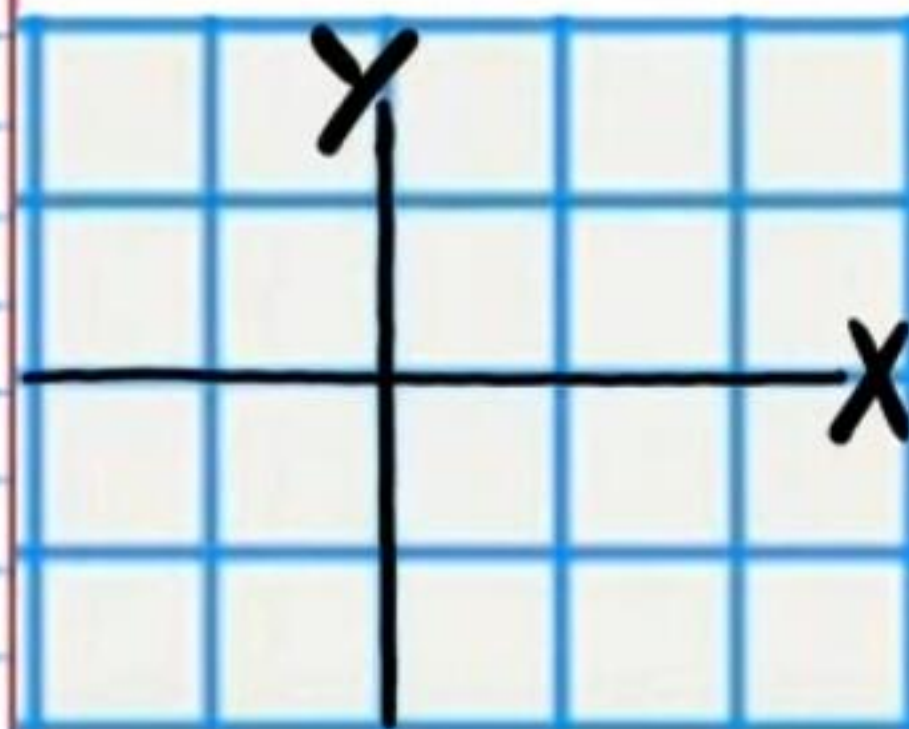


23) Graph the following equation using radian units. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

$$r = 2\sqrt{\cos(2\theta)}$$

1st Point

2nd Point

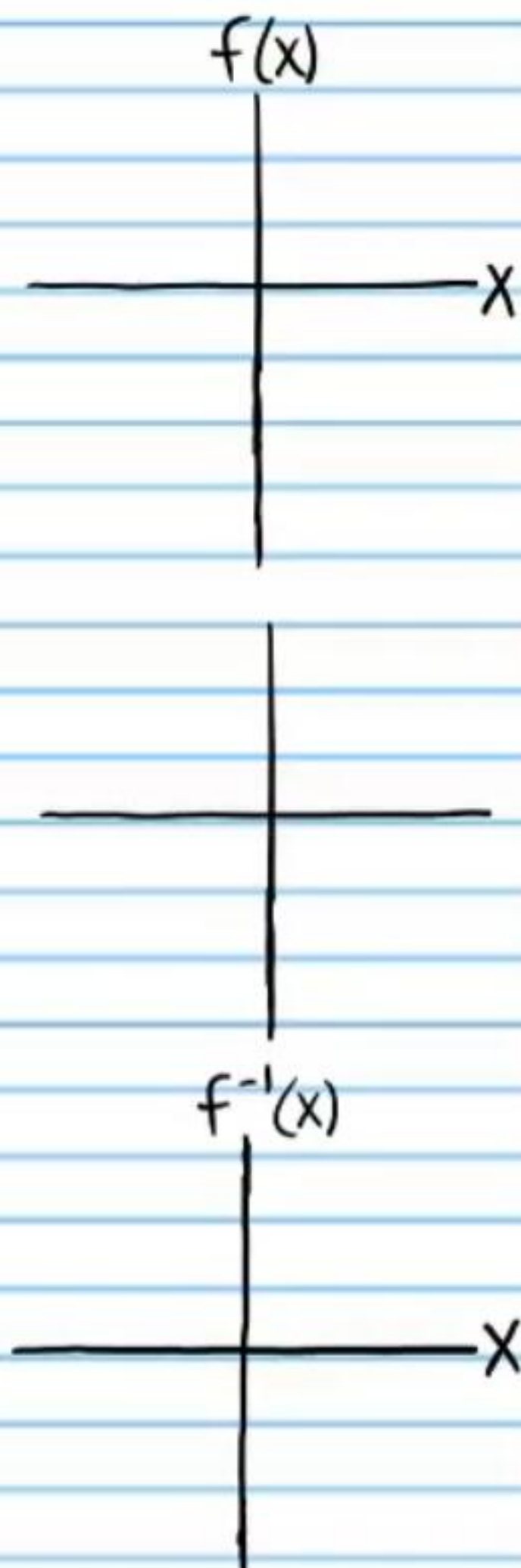


3rd Point

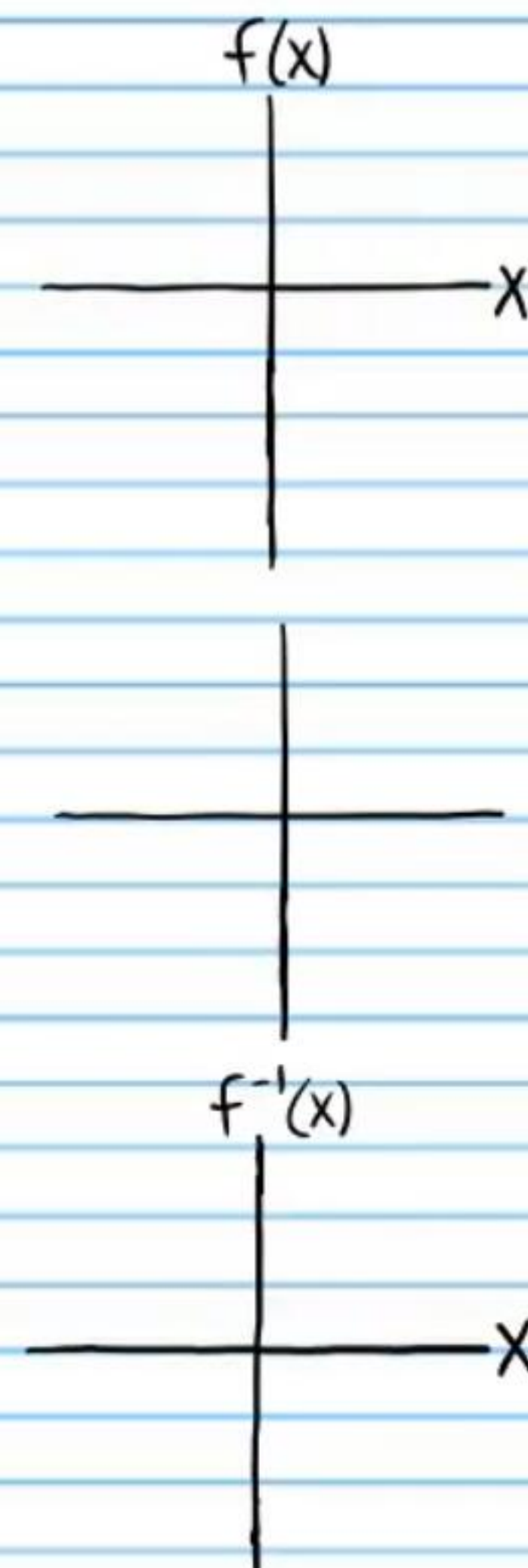
4th Point

For problem #'s 24-25, find the equation of the inverse function. Restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

24) $f(x) = \sqrt{x+7}$



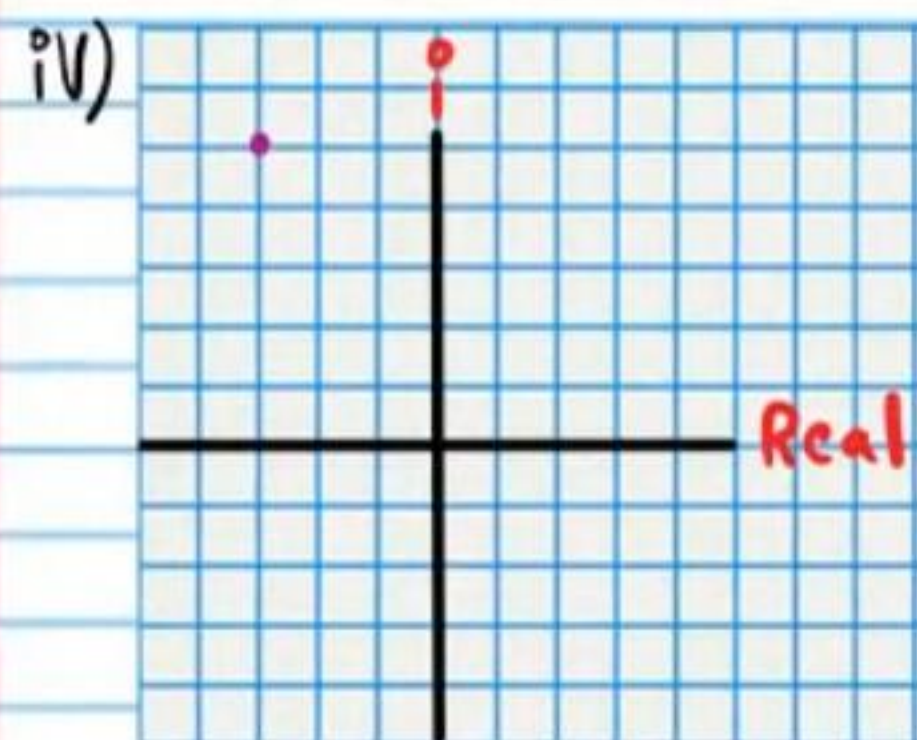
25) $f(x) = \tan x$



Practice Test IV (Version 2) Answers

① 25.47 days

② i) $6,000 \text{ E. coli}$ ii) 133 E. coli iii) $1,322 \text{ E. coli}$



③ $x \approx 1.099$

④ $x = \frac{1}{\sqrt[3]{10,000}}$ or $x = \frac{1}{(\sqrt[3]{10})^4}$

⑤ 13.5

* Notice that this number is more than the number of people he actually met. It is the speed that he met people, not the number of people he met.

⑥ i) 2.5 people/day ii) 15 people/day

⑦ $DQ = \frac{1-3(x+h)-(1-3x)}{h}$

⑧ $17.89 \text{ grams titanium-45}$

⑨ $10^{-8.9} \text{ W/m}^2$ or $1.2589 \times 10^{-9} \text{ W/m}^2$

⑩ i) $b=0 \text{ or } b=4$ ii) $k=240$

⑪ $x \approx 54.598$

⑫ $x = \frac{-2 \pm \sqrt{19}}{3}$

⑬ i)

X	11	10	18	4	3
f(x)	16	16	16	3	4

X	16	16	16	3	4
f ⁻¹ (x)	11	10	18	4	3

Omit any two of these points.

ii)

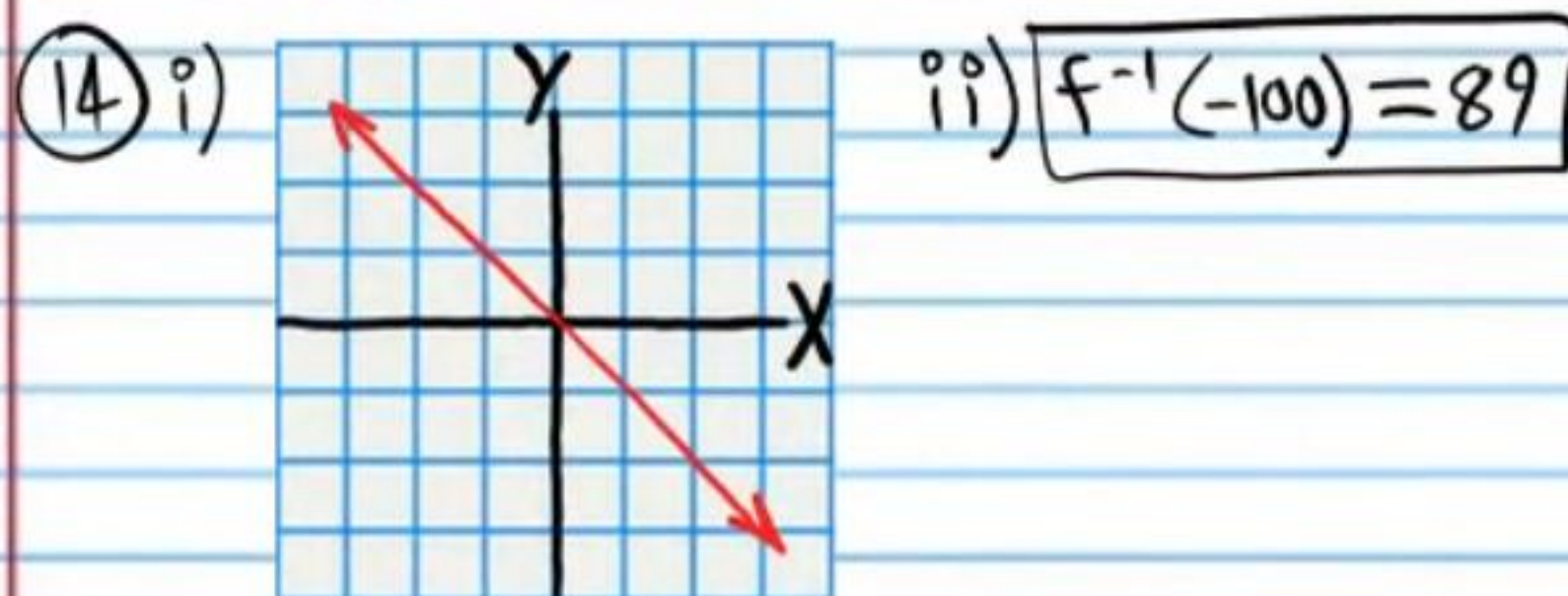
X	-2	-4	-5	-7	8
f(x)	-3	-1	9	-7	9

X	-3	-1	9	-7	9
f ⁻¹ (x)	-2	-4	-5	-7	8

Omit any one of these points.

iii) $\log_9 12 = \frac{\ln 12}{\ln 9} \approx 1.1309$

iv) $g(-9) = 4$



⑮ 0.99 years

⑯ $f(g(x)) = \sqrt[3]{(x-1)^3 + 2} - 2 + 1 = \sqrt[3]{(x-1)^3} + 1 = x - 1 + 1 = x$ ✓

$g(f(x)) = (\sqrt[3]{x-2} + 1 - 1)^3 + 2 = (\sqrt[3]{x-2})^3 + 2 = x - 2 + 2 = x$ ✓

Yes

①7 $\boxed{-14}$

①8 $\boxed{x = \pm 4\sqrt{19} - 20}$

①9 $\boxed{f^{-1}(x) = \frac{2}{x}}$

②0 $\boxed{f^{-1}(x) = \log(-\frac{x-4}{3})}$

②1 $f(f^{-1}(x)) = \frac{3}{5(\frac{3}{5x} - \frac{2}{5}) - 2} = \frac{3}{\frac{3}{x} - 2 + 2} = \frac{3}{\frac{3}{x}} = x \checkmark$

$$f^{-1}(f(x)) = \frac{3}{5(\frac{3}{5x+2})} - \frac{2}{5} = \frac{3}{\frac{15}{5x+2}} - \frac{2}{5} = \frac{15x+6}{15} - \frac{2}{5}$$

$$= \frac{5x+2}{5} - \frac{2}{5}$$

$$= \frac{5x}{5}$$

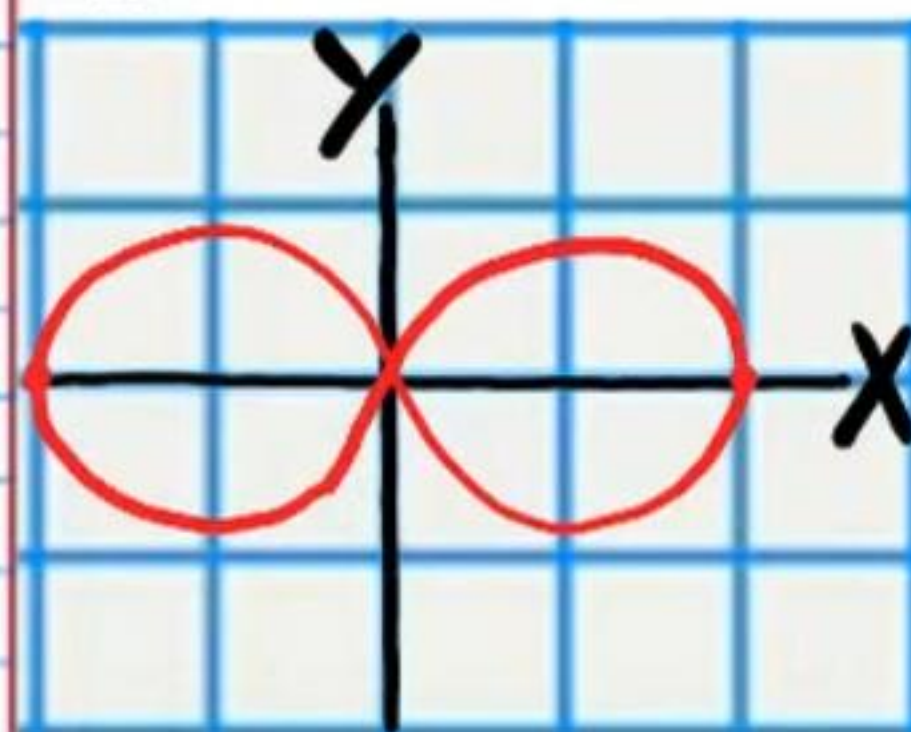
$= x \checkmark$

②2 i) $\boxed{\text{Not a Function}}$ ii) $\boxed{\text{Function}}$ iii) $\boxed{\text{Function}}$

iv) $\boxed{1\text{-to-1 Function}}$ v) $\boxed{1\text{-to-1 Function}}$ vi) $\boxed{\text{Function}}$

vii) $\boxed{\text{Function}}$ viii) $\boxed{\text{Not a Function}}$

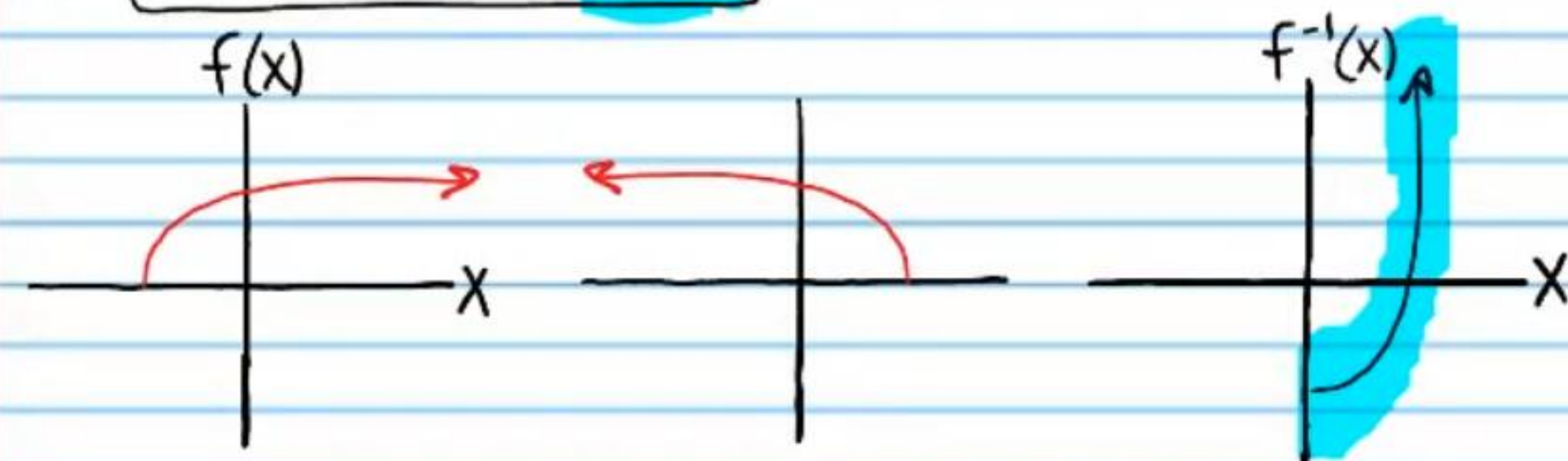
②3 $\star$ Other points are possible. Only 4 are necessary.



$(0, \frac{\pi}{4})$ $(2, 0)$

$(2, \pi)$ $(\sqrt{2}, \frac{5\pi}{6})$

②4 $\boxed{f^{-1}(x) = x^2 - 7, x \geq 0}$



②5 $f(x) = \tan x, \boxed{-90^\circ < x < 90^\circ}$

$\boxed{f^{-1}(x) = \tan^{-1} x}$

$-90^\circ < y < 90^\circ$ You may write the range restriction, but it is implied so is not necessary.

